Pilz PSEN cs3.1 RFID Non-Contact Safety Interlock Switch Manual

Safety Category: EN ISO 13849-1 Category 4, Performance Level e (PL e); IEC 62061 SIL CL 3.

Technology: Coded RFID non-contact transponder technology offering high tamper resistance.

Operating Distances: Assured release distance Sao = 15 mm, Assured unlatch distance Sar = 40 mm.

Typical Application: Perimeter barrier monitoring on Safety Gate 2 guarding Cell A robot envelope.

Housing Rating: IP67 washdown-resistant enclosure suitable for manufacturing cells.

Pilz PSEN cs3.1 - Dual-Channel Wiring and Relay Interfacing

Wiring per Schematic Drawing E102: Output OSSDs connect to safety inputs I:0/10 and I:0/11.

Terminal TB1-21 connects Channel 1 (OSSD1); Terminal TB1-22 connects Channel 2 (OSSD2).

Diagnostic LEDs on Sensor Head: Solid Green = Both channels closed and safe; Red = Safety circuit open.

Flashing Red Diagnostic Code: Indicates channel discrepancy timeout or cross-circuit wiring fault.

Emergency Stop Relay K1 monitors gate contacts and cuts 400V drive power upon interlock opening.

Pilz Safety Systems - Reset Protocols and Alarm Recovery

When Safety Gate 2 opens during machine cycle, alarm ALM-205 Safety Gate Open is latched.

Recovery Sequence: Close physical gate, ensure actuator tongue is seated, press blue Safety Reset.

Safety relay K1 re-energizes main servo contactors and clears passivated state on PLC01 F-DI modules.

Periodic Maintenance: Perform functional trip test of interlock switch every 90 calendar days.